CSE 4345: Big Data Analytics

Week 9, Part 1 — Data Visualization: Principles, Techniques & Big Data Tools

Department of Computer Science & Engineering
Premier University, Chittagong

Semester 2026

Course & Lecture Information

Lecture Identity

Course:	CSE 4345
Week / Part:	9 of 11 | Part 1 of 2
CLO:	CLO3
PLO:	PLO(e)

CLO3

"Explain the importance and performance of different big data techniques and tools used to store, manage, and analyse large datasets"

Course Progression

Week 7: Hive, HiveQL, SQL-on-Hadoop
Week 8: Kafka, Sqoop, Flume, Lambda/Kappa

Part 1 Covers

- Why visualization is a critical skill in big data analytics
- Tufte's principles: data-ink ratio, small multiples, sparklines
- Shneiderman's visualization mantra and seven task taxonomy
- Choosing the right chart by data type
- Big data visualization challenge 1: overplotting (hexbin, density contour)
- Big data visualization challenge 2: latency (pre-aggregation, data cubes)
- Big data visualization challenge 3: streaming (real-time chart updates)
- Tools landscape: Tableau, Power BI, D3.js,

Lecture Agenda

Why Visualization Matters in Big Data

From Data to Insight: The Visualization Imperative

Anscombe's Quartet (1973)

Four datasets with **identical summary statistics** (mean, variance, correlation $r = 0.816$) that look completely different when plotted. This 50-year-old demonstration remains the canonical argument for always visualising data before trusting statistics.

	All 4 datasets
Mean of x	9.0
Mean of y	7.5
Variance of x	11.0
Correlation r	0.816
Linear fit	$y = 3 + 0.5x$

Yet Dataset I is linear, II is curved, III has an outlier

The Big Data Dimension

At petabyte scale, summary statistics are even more dangerous because:

- A single table with 10 billion rows may contain multiple overlapping populations with opposite trends that average out
- Log-scale distributions (income, city size, web traffic) make arithmetic means meaningless
- Streaming data can hide sudden regime changes that a daily average obscures

Visualization is the first and most powerful quality check on any big data analysis.

Bangladesh Example

DGHS monthly disease count averages look stable

Tufte’s Principles of Analytical Design

Tufte: Maximise the Data-Ink Ratio

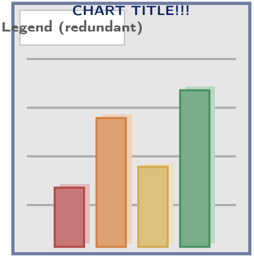
The Three Core Principles (Tufte, 1983)

1. **Data-ink ratio:** Of all the ink used in a chart, maximise the fraction that encodes actual data. Eliminate "chart junk" — decorative borders, 3D effects, gradient fills, redundant legends, unnecessary gridlines.

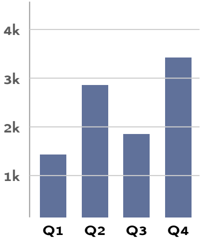
$$\text{Data-ink ratio} = \frac{\text{ink encoding data}}{\text{total ink in chart}}$$

2. **Small multiples:** Repeat the *same* chart structure across different subsets (districts, years, categories) for direct visual comparison. The eye detects differences and trends across a grid of small charts far more quickly than by reading separate tables.

Before (low data-ink ratio)



After (high data-ink ratio



Rule

If erasing an element does not reduce the information in the chart, erase it.

Tufte: Small Multiples and Sparklines

Small Multiples

A small multiple (also called a **trellis** or **facet** chart) repeats the same visual structure across partitions of the data. The result is a grid where the eye can instantly compare trends:

same scale: cross-district comparison is valid

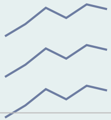
Each panel uses the **same axis scale** so relative magnitudes are directly comparable.

Sparklines

Sparklines embed a trend line directly inside a table or text, eliminating the need for a separate figure. They communicate direction and shape without requiring the reader to track a separate chart.

District Daily Case Count

Dhaka	4,217	
Ctg	1,803	
Khulna	892	



sparkline = trend in-line with the number

Shneiderman's Visualization Mantra and Taxonomy

The Visualization Mantra (Shneiderman, 1996)

"Overview first,
zoom and filter,
then details on demand."

This three-step interaction model applies to **every** interactive visualization system. A user should always be able to:

1. See the **whole picture** first (global context)
2. **Narrow down** to the region or subset of interest (zoom/filter)
3. Retrieve the **raw record** for any item when needed (drill-down)

This model directly shapes modern BI tools: Tableau's

Shneiderman's Seven Data Types

Data Type	Best Chart Types
1D Linear	Bar chart, histogram
2D Map	Choropleth, bubble map
3D World	3D scatter, volume render
Temporal	Line chart, area, calendar heatmap
Multi-dim	Scatterplot matrix, parallel coords, heatmap
Tree	Treemap, sunburst, dendogram

Five

Choosing the Right Chart

Visualization Selection by Data Type and Question

Data Type	Analytical Question	Recommended Visualization	Avoid
Temporal	How did X change over time?	Line chart (trend), area chart (cumulative), calendar heatmap (daily pattern)	Pie chart
Categorical	Which category is largest?	Horizontal bar chart (>5 cats); grouped bar (comparison across groups)	3D pie chart
Geographical	How is X distributed across regions?	Choropleth (rate/proportion); proportional bubble map (absolute count)	Plain bar for spatial data
Relational	How are entities connected?	Force-directed network graph; chord diagram (flow between groups)	Table
Distribution	What is the spread and shape of X?	Histogram (shape); box plot (quartiles + outliers); violin (density +	Line chart

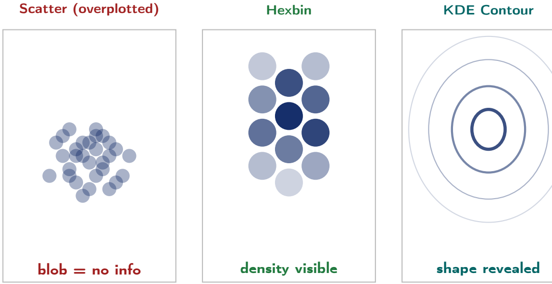
Big Data Visualization Challenges

Challenge 1: Overplotting at Scale

The Overplotting Problem

A standard scatterplot renders one point per data record. At **millions of records**, points overlap completely — the chart becomes a solid blob of ink that conveys zero information beyond "data exists here." **Three solutions in increasing sophistication:**

1. **Alpha transparency (opacity):** Set each point to $\alpha = 0.02$. Denser regions appear darker. Simple but still fails at very high densities.
2. **Hexbin aggregation:** Divide the 2D plot area into a hexagonal grid. Count points in each hex. Colour each hex by count. A scatter of 10 million points becomes a grid of $\sim$10,000 hex cells — interactive and readable.

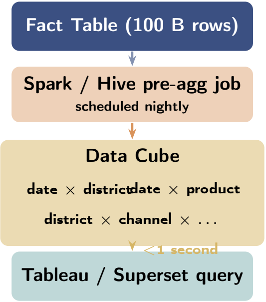


Challenge 2: Latency and Pre-Aggregation

The Interactivity Problem

A business analyst using Tableau expects a dashboard filter (e.g., *“show Q1 2024 only”*) to respond in under **1 second**. But the underlying table has **100 billion rows** in Hive on HDFS. A Hive query over 100 billion rows takes minutes, not milliseconds. **Two solutions:**

1. **Pre-aggregation with a data cube:** Materialise aggregate tables at all combinations of dimensions (district × date × product × channel) in advance. Queries read from the pre-computed cube (millions of rows) rather than the raw fact table (billions of rows).
2. **Approximate query processing (AQP):** Return a statistically correct estimate of the aggregate in milliseconds by sampling the table. Tools:



cube size: millions of rows (not billions)

trade-off: staleness (up to 24 h) for speed

Challenge 3: Streaming Visualization

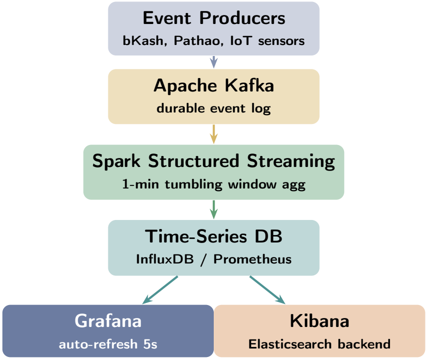
Real-Time Chart Updates

Static charts describe the past. Many big data applications need **continuously updating visualizations**:

- Pathao operations dashboard: live driver count by district, updated every 5 seconds
- bKash fraud operations: transaction volume per minute, alert rate, active case count
- DGHS disease surveillance: daily case counts updating as hospitals submit reports

Streaming visualization architecture:

1. Events arrive at **Kafka** topic in real time
2. **Spark Structured Streaming** or **Flink** aggregates over a sliding window (e.g., 1-minute tumbling window)



WebSocket vs. Poll

Grafana polls the TSDB on a schedule (5s refresh). For sub-second updates (stock tickers,

Visualization Tools Landscape

Tools Landscape: Choosing the Right Tool

Tool	Best For	Data Source	Scale	Real-time
Tableau	Business dashboards, drag-and-drop BI	Spreadsheets, SQL, Spark, Hive	Millions of rows	Limited
Power BI	Microsoft ecosystem, executive reports	Azure, Excel, SQL Server, REST API	Millions of rows	Limited
D3.js	Custom interactive web visualizations	JSON, CSV, REST APIs	Flexible (client-side)	Yes (WebSocket)
Grafana	Real-time metrics, ops dashboards	InfluxDB, Prometheus, Elasticsearch	Millions of data points	Yes (5s poll)
Apache Superset	Open-source BI on data lakes	Presto, Hive, Spark SQL, PostgreSQL	Billions of rows	Limited

Ivy League Alignment

Data Visualization in Top University Curricula

Harvard CS171 — Visualization (Pfister & Borkin)

Harvard's dedicated visualization course, widely regarded as the gold standard:

- **Week 1** opens with Tufte's data-ink ratio and Anscombe's Quartet as the canonical argument for visualization
- **D3.js** is the mandatory implementation framework — all assignments are interactive SVG charts in the browser
- **Shneiderman's mantra** is the grading rubric for interaction design: students lose marks if their dashboard does not implement overview → zoom → detail
- **Streaming data lab**: students wire a Kafka consumer to a D3.js chart via WebSocket and visualise Twitter sentiment in real time at 1,000 tweets/minute
- CS171 final projects have been published on the BBC

Stanford CS246 — MMDs (Leskovec)

Stanford uses visualization as an **analysis tool**, not just a presentation tool:

- t-SNE and UMAP projections of high-dimensional graph embeddings help students validate whether their community detection algorithms produce meaningful clusters
- Choropleth maps of PageRank scores over a geographic web graph reveal that high-authority pages cluster geographically — an unexpected analytical insight discovered through visualization

Student Assessment Pool

Assessment—MCQ Set A

CLO3

Instructions

Select the single best answer. Correct answers **bolded** for instructor reference.

Q1. Edward Tufte's "data-ink ratio" principle states that a visualization should:

- (a) Use more colours to highlight different categories of data
- (b) **Maximise the fraction of ink devoted to representing actual data, removing all decorative elements**
- (c) Use ink sparingly to reduce printing costs
- (d) Always include data labels using a different ink colour from the chart body

Q2. Which visualization type is most appropriate for showing the **distribution and spread** of a continuous variable?

- (a) Pie chart (b) Line chart (c) **Histogram** (d) Treemap

Instructor Notes

Q1: Option (c) is a classic distractor — the data-ink ratio is about what the ink *encodes*, not how much is used. A chart with one thick bar encoding one important value has a high data-ink ratio. **Q2:** A pie chart (a) shows part-to-whole; a line chart (b) shows change over time; a treemap (d) shows

Q3. Shneiderman's visualization mantra begins with which step?

- (a) Details first, then overview
- (b) **Overview first, zoom and filter, then details on demand**
- (c) Filter first, then provide an overview
- (d) Details on demand first, then zoom out

Q4. Which visualization tool is **best suited** for a real-time production operations dashboard monitoring server metrics and application error rates with 5-second refresh intervals?

- (a) Tableau (b) D3.js (c) Power BI (d) **Grafana**

Instructor Notes

Q3: Options (a), (c), and (d) invert the mantra. Students who select (c) "filter first" have confused the interaction sequence — filtering is only meaningful after the user has seen the overview and identified what to filter. **Q4:** Tableau (a) and Power BI (c) support limited scheduled refresh but are not designed for second-level streaming metrics. D3.js (b) can do real-time with WebSockets but requires custom development; Grafana (d) natively integrates with time-series databases (InfluxDB, Prometheus) for live dashboards.

Assessment—True / False *CLO3*

Instructions

State True or False and provide a **one-sentence justification** for full marks.

Statement	**Answer**
1. A pie chart is a recommended choice for comparing more than 5 categories.	**False**
2. Hexbin plots are an effective technique for reducing over-plotting when visualising millions of data points.	**True**
3. D3.js generates static PNG or PDF image files as its primary output format.	**False**
4. Choropleth maps are used to visualise the geographic distribution of a quantitative variable.	**True**

Assessment—Descriptive Questions *CLO3*

Instructions

Answer each question in **100–150 words**. Include diagrams where indicated.

D1. Explain Tufte's **data-ink ratio** principle. Provide a "before and after" example: describe a cluttered bar chart (what chart-junk elements are present) and then describe the clean version after applying the principle.

D2. What is **overplotting**? Describe **three techniques** to address it when visualising millions of data points in a scatterplot. For each technique, explain what information it preserves that the original overplotted chart lost.

D3. What is a **data cube**? How does pre-aggregating a billion-row fact table into a data cube enable interactive visualization at sub-second latency? What is the trade-off of this approach?

Instructions

Answer in **200–300 words** with step-by-step reasoning. Reference Shneiderman's taxonomy and Tufte's principles where applicable.

A1. Public Health Dashboard Design:

A DGHS public health dashboard needs to communicate:

- (a) COVID-19 confirmed case counts by district over 24 months
- (b) Real-time hospitalisation rate updated every 5 minutes
- (c) Demographic breakdown (age group, gender) of confirmed cases

For each sub-requirement: specify the appropriate visualization type, justify using Shneiderman's data-type taxonomy, identify which big data visualization challenge applies (if any), and name the tool you would use. Apply Tufte's data-ink principle to state one specific chart-junk element to eliminate.

A2. Streaming Sentiment Visualization:

Design a complete visualization system for a streaming Twitter (X) sentiment analysis pipeline processing **10,000 tweets per second**. Specify: (a) which Kafka aggregation window you would use and why; (b) which chart types best convey sentiment drift over time and topic clustering; (c)

Textbook & Reading References

Primary Textbooks for Week 9

- **[SA]** Acharya & Chellappan. *Big Data and Analytics*, Wiley, 2015. **Ch. 10**
  - Ch. 10: Data Visualization — chart types, big data visualization tools, dashboard design
- **[TE]** Erl, Khattak & Buhler. *Big Data Fundamentals*, Prentice Hall, 2016. **Ch. 12**
  - Ch. 12: Big Data Analysis and Visualization — analytical techniques, output communication

Seminal Works (covered in Week 9, Part 2)

- Tufte, E. R. (1983). ***The Visual Display of Quantitative Information.*** Graphics Press. [Required at Harvard, Yale, Stanford — the founding text of analytical visualization]
- Shneiderman, B. (1996). **The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations.** *IEEE Symposium on Visual Languages.* [Defines the overview-zoom-filter-details mantra and seven data types]
- Bostock, M., Ogievetsky, V., & Heer, J. (2011). **D^3: Data-Driven Documents.** *IEEE Transactions on Visualization and Computer Graphics.* [Introduces D3.js; most cited visualization systems paper of the 2010s]

Questions & Discussion

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"Above all else, show the data."

— Edward R. Tufte, *The Visual Display of Quantitative Information*, 1983

Next Lecture: Week 9, Part 2

Seminal Papers: Tufte (1983) · Shneiderman (1996) · Bostock et al. D^3 (2011)

Case Study: **New York Times Graphics Desk** — D3.js real-time election visualizations, communicating uncertainty with probability distributions, live data streaming as results arrive on election night

Reading before Part 2: [SA] Ch. 10 · Tufte (1983) Ch. 1–2 (data-ink ratio, chart junk)